Reusing witnesses in Diophantine consistency equations

Deterministic CRT moduli, cofinal horizons, and shorter Pell transport

AI-assisted research prepared for Vladimir Reshetnikov
ProveIt research program

24 October 2026

Abstract

This packet gives explicit ordinary integer polynomials representing finite refutability of ZFC, with complete coefficient data and ordinary proofs of the new transformations. The principal new idea is to reuse a power already certified by a proof-certificate construction as the size bound for its Chinese remainder encoding. A freely chosen encoding modulus can then be replaced by a deterministic polynomial in existing coordinates. Independently, the arbitrary padded proof horizon can be restricted to a cofinal family of squares, and the sign of a proposed factorial output can be recovered from existing power and ceiling equations. Choosing the modulus as a product of consecutive integers makes its sign unconditional. The principal result is an explicit quartic in 508 unrestricted integers with 6,545 monomials and coefficient height 996,521, improving the matched 517-variable parent in variables, residuals, terms, and height. A separate positive-domain construction has ten unknowns, degree 288, and an 89-operation evaluator. A simpler 512-variable checkpoint and five degree/cost companions are retained. The direct construction encodes the numerical proof calculus C_{222}; it does not simulate K215 or instantiate Jones's universal polynomial. All claims of improvement are relative to the frozen ProveIt sources identified below. No global minimum, literature-wide priority, proof of consistency, or new end-to-end proof-assistant verification is asserted.

Contents

1 Results, conventions, and comparison boundary **2**

2 The direct proof-certificate interface **3**

3 Cofinal witness restriction **4**

4 A deterministic CRT modulus from an existing power **5**

5 The complete 512-variable integer quartic **6**

6 A signed factorial output: the 508-variable refinement **7**

7 Ten positive unknowns with an 89-operation evaluator **10**

8 Verification, reproducibility, and evidence boundaries **13**

9 Further research questions **14**

A Complete quadratic residuals of the principal equation **16**

1 Results, conventions, and comparison boundary

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$. Unless a result explicitly says "strictly positive", every unknown ranges independently over $\mathbb{Z}$. A polynomial abbreviation is an instruction to substitute an integer polynomial; it introduces no quantified unknown. Our equations contain no variable exponent, factorial operation, division, square-root operation, or unspecified coefficient in their final arithmetic representations.

The repository was frozen at commit

`45b399ec51dfbecb891dd51d8a9e4b6b7c361ac6`.

Both the active candidates and the eight archives then present in `docs/incoming` were examined. This matters because the overview of the older manuscript collection is behind the most recent incoming constructions. In particular, the matched direct parent is the 517-variable integer quartic in the incoming Round 14 packet, rather than an older 569-, 604-, or 663-variable construction [2, 1].

Integer quartic	Variables	Rows	Terms	Height
Matched incoming parent	517	310	6,774	3,904,267
Cofinal/modulus checkpoint	512	310	6,742	3,904,268
Signed-factorial principal	**508**	**308**	**6,545**	**996,521**
Incoming lower-height alternative	519	321	6,253	411,060

All three have exact degree four. The 512-variable checkpoint slightly increases height; the principal 508-variable equation improves every displayed quantity relative to the matched parent. A separate family has ten strictly positive unknowns, with operation/degree pairs 89/288, 91/283, 92/278, 94/262, 96/258, and 97/254. Section 7 specifies those polynomials and their domains. The separate 519-variable incoming equation retains fewer terms and smaller coefficients than the 508-variable result; it remains a useful tradeoff.

Here "rows" means integral quadratic polynomials R_j in the representation $P = \sum_j R_j^2$, and coefficient height means the largest absolute coefficient after the full sum has been expanded and like monomials collected. Arithmetic operation counts for the compact equations count each live binary addition, subtraction, and multiplication, including squares and multiplication by a fixed constant. They do not measure bit complexity: those equations have very large fixed native coefficients.

There is no single established meaning of "simplest" that orders all these equations. A few unknowns, low degree, small coefficients, few terms, short evaluation circuits, and short correctness proofs are different objectives. The direct quartics improve a modest-coefficient, independently encoded branch of the repository. They do not supersede the repository's separate 37-integer quartic with large native coefficients, its ten-integer higher-degree constructions, or all of its positive-domain forms.

1.1 What is proved in ZFC

Fix the finite syntactic ZFC-refutation relation of the retained source calibration and write

$$\mathrm{Con}(\mathrm{ZFC}) \ :\Longleftrightarrow\ \neg\exists p \in \mathbb{N}\ \mathrm{Prf}_{\mathrm{ZFC}}(p, \bot).$$

For each asserted equation the target is the implication in both directions,

$$\mathrm{ZFC} \vdash (\exists \mathbf{z} \in D^n\ P(\mathbf{z}) = 0) \longleftrightarrow (\exists p \in \mathbb{N}\ \mathrm{Prf}_{\mathrm{ZFC}}(p, \bot))\,, \tag{1}$$

where D is the stated domain. Negation gives the requested consistency equivalence. The proof is an ordinary mathematical proof formalizable in ZFC, composed with precisely identified inherited calculus and arithmetic theorems. No truth or soundness assumption about ZFC is used to infer a refutation from a polynomial zero. Conversely, a finite syntactic refutation is used as finite data for constructing integer witnesses.

The artifact checks establish literal polynomial identities, coordinate maps, coefficient expansions, and numerical metrics. They do not establish unsolvability, decide consistency, or replace the unbounded arithmetic proofs. The supplied Lean interface files document the inherited calibration; their entire transitive import closure has not been rebuilt for this packet.

2 The direct proof-certificate interface

The direct construction begins with the repository's numerical calculus C_{222}. Its implication, membership, universal-quantifier, and equality constructors are fixed integer polynomials in the schema parameters; it has finitely many axiom and inference branches, a truth code 0, and a falsity code 1. The complete constructor formulas and the guarded axiom condition are retained with the source documents. In the square-shell variant used here, the corresponding syntax is related to the original coding by the retained finite syntax-transfer argument; it is not an arbitrary relabeling.

The inherited logical calibration is

$$C_{222} \text{ derives } 1 \iff \text{ZF has a finite refutation} \iff \text{ZFC has a finite refutation.} \tag{2}$$

The named interfaces are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The second uses the constructible-universe relative consistency argument. The filename of the first interface does not make the present arithmetic construction a simulation of a Turing machine.

2.1 The parent theorem used by the new reduction

The retained direct theorem has the following concrete properties.

1. A finite refutation can be expanded to a finite proof tree. After choosing any sufficiently large integer horizon $t \geq 16$, truth steps pad the backward proof expansion to exactly t rows. The lower bound on t depends only on finitely many proof-tree parameters and constructor labels, including the fixed truth-padding tuple.

2. The semantic factorial is $F = t!$, and the label of code j is assigned the positive weight $W_j = 1 + jF$ for $0 \leq j \leq t$.

3. A backward expansion uses positive accumulators S_i and S_{i+1}, which are products of the weights of the currently pending formulas. A row with conclusion v and premises p, q satisfies

$$S_{i+1} W_v = S_i W_p W_q. \tag{3}$$

   Unused premises have code 0, whose weight is 1. The endpoints are $S_0 = F + 1$ and $S_t = 1$.

4. The first CRT encoding uses moduli $1 + (i+1)y$, $0 \leq i \leq t$, where y is any sufficiently large positive multiple of F. In particular, $S_i \leq y$ and $y \geq F + 1$ suffice. The raw endpoint slack is $r_0 = y - F - 1 \geq 0$.

5. The factorial module already certifies the power

$$\rho = Ft^2, \qquad V = (\rho + 1)^t. \tag{4}$$

   Its noncircular signed-power and digit argument recovers $F = t!$ and the intended value of V from every parent zero.

6. A second, prime-power CRT encoding and shared Pell construction compile the bounded local conditions into a finite integral quadratic system. The integer version imposes a finite selected list of nonnegativity conditions with private three-variable representations. Exact monic elimination and integer translation produce the delivered 517-variable quartic.

These are the actual completeness and soundness properties proved in `FACTORIAL_CARRIER_PROOF.md`, `DERIVED_LABEL_RANGE_PROOF.md`, `HYBRID_PROOF.md`, and the retained signed-source proof. They are included in the packet, together with the literal parent equation and its source ledger. The new argument does not replace any of these statements by an asymptotic assertion about an unspecified universal polynomial.

2.2 Why the multiplicative flow is sound

The positive weights $W_1, \dots, W_t$ are pairwise coprime. Indeed, a prime dividing both W_j and W_k would divide $(j-k)F$ while not dividing F. It would therefore be at most t, hence would divide $F = t!$, a contradiction. Multiplying (3), cancelling the overlapping positive accumulators, and taking a prime valuation private to each W_j proves

$$\#\{\text{conclusions with label } j\} - \#\{\text{premises with label } j\} = \begin{cases} 1 & j = 1, \\ 0 & j > 1. \end{cases} \tag{5}$$

Cycles in the rule multiset cause no gap. Let D be the closure of truth under its finitely many valid rule instances. If falsity were outside D, each rule concluding outside D would have a premise outside D. The number of outside conclusions would be at most the number of outside premises, contradicting the sum of (5) over labels outside D. Thus falsity has a finite derivation. This is the logical reason the compact unordered certificate still represents refutability.

3 Cofinal witness restriction

The new restriction is applied to a freely chosen part of a completeness construction. It is not a surjectivity claim for squares.

Lemma 3.1 (Cofinal selection)**.** *Suppose that every finite certificate c has realizations with every integer parameter $t \geq L(c)$, with all later witnesses chosen after t. Let $\phi : \mathbb{Z} \to \mathbb{N}$ have unbounded image and let the extra equation $t = \phi(u)$ be expressible by the permitted polynomial system. If the original system is sound for every such t, the restricted system has a zero if and only if a finite certificate exists.*

Proof. A restricted zero is an original zero. Conversely choose u with $\phi(u) \geq L(c)$ and then carry out the original completeness construction at that parameter. The order of choices is essential: witnesses depending on t are regenerated after u has been selected. □

For the present certificate, take

$$t = 16 + u^2. \tag{6}$$

There is no congruence condition on the padded proof horizon. Given the finite proof tree, choose $|u|$ large enough to dominate its step count, every local schema parameter, and every complete constructor-label bound. Padding by truth remains available. The factorial, weights, accumulators, both CRT encodings, and Pell witnesses are then constructed afresh.

The original unrestricted-integer implementation used

$$t - 16 = a^2 + b^2 + c(c+1).$$

Replacing it by (6) removes two private coordinates and keeps the row quadratic. The classical equivalence

$$n \geq 0 \quad \Longleftrightarrow \quad \exists a, b, c \in \mathbb{Z}\ n = a^2 + b^2 + c(c+1) \tag{7}$$

follows from the three-square theorem applied to $4n+1$; see also the explicit formulation in [5, Lemma 6.1]. The reverse implication follows directly from $c(c+1) \geq 0$ for integral c.

Remark 3.2 (The restriction is not valid for an arbitrary coordinate). An equation forcing a natural witness to equal 2 has a solution, whereas forcing that witness to be a square destroys all its solutions. Similarly, an unbounded admissible set such as $\{4k+2 : k \in \mathbb{N}\}$ contains no square. Unboundedness alone is insufficient. We use the parent's stronger fact that every sufficiently large horizon is admissible, with the other witnesses chosen afterwards.

4 A deterministic CRT modulus from an existing power

Lemma 4.1 (Pending-product bound). *Let a proof tree be expanded backwards for at most t steps, with each step replacing one pending conclusion by at most two premises. Suppose all labels lie in $[0, t]$, $t \geq 1$, and $F \geq 1$. Put $A = 1 + Ft^2$ and $V = A^t$. Every accumulator in the expansion, including truth padding, is at most V^2.*

Proof. The initial pending multiset has one member. After s steps its cardinality is at most $s+1$, hence at most $t+1$. Each factor is $W_j = 1 + jF \leq 1 + tF \leq A$. Consequently

$$S_i \leq A^{t+1} \leq A^{2t} = V^2.$$

Truth has weight 1 and leaves this bound unchanged. □

The expression V^2 already belongs to the same arithmetic certificate: the parent does not need a new exponentiation request to obtain it. Moreover,

$$V = (1 + Ft^2)^t \equiv 1 \pmod{F}. \tag{8}$$

This congruence permits a deterministic modulus with both a size property and the necessary divisibility property.

Proposition 4.2 (A family of polynomial moduli). *Let integers $\kappa \geq 2$ and $d \geq 1$ satisfy $(\kappa - 1)d \geq 2$. Replace the freely chosen modulus and its slack by*

$$F \geq d, \qquad y = V^2 + \kappa F - 1. \tag{9}$$

For arbitrary signed V, these conditions imply $r_0 = y - F - 1 = V^2 + (\kappa - 1)F - 2 \geq 0$. For canonical $F = t!$ and V from (4), they also imply $F \mid y$ and $y > V^2$. Thus they supply every first-CRT condition required by the parent completeness construction.

Proof. The slack inequality is immediate from the displayed formula and the condition on d. Congruence (8) gives $V^2 + \kappa F - 1 \equiv 0 \pmod{F}$. Finally $\kappa F - 1 \geq 1$, so the pending-product lemma implies every required accumulator is strictly smaller than y. □

The delivered 512-variable equation uses $\kappa = 3, d = 1$:

$$\boxed{y = V^2 + 3F - 1, \qquad F \geq 1.} \tag{10}$$

Strengthening the existing $F \geq 0$ row to $F \geq 1$ changes a constant but introduces no coordinate. The former three private witnesses for $r_0 \geq 0$ are deleted. The new modulus equation is a single quadratic row. There is no requirement to store V^2 as another unknown.

4.1 The CRT construction remains valid

For completeness, $F = t! \mid y$. Suppose a prime p divides both $m_i = 1 + (i+1)y$ and $m_j = 1 + (j+1)y$, where $0 \leq i < j \leq t$. It does not divide y, so $p \mid j - i$ and $p \leq t$. But then $p \mid F \mid y$, a contradiction. Therefore these moduli are pairwise coprime. The Chinese remainder theorem supplies a nonnegative x with $x \equiv S_i \pmod{m_i}$ for all i. As $0 < S_i \leq y < m_i$, the corresponding Euclidean quotients are nonnegative. The initial and final residues are exactly $F + 1$ and 1.

The order of the construction is

$$\text{finite proof} \longrightarrow t \longrightarrow F, V, y \longrightarrow x \longrightarrow \text{second CRT and Pell witnesses.}$$

In particular, the deterministic choice of y has no dependence on the later code x. The second CRT construction still chooses its representatives at the original strengthened prime-power precision. No claim that one fixed small representative meets all amplified congruences is introduced.

5 The complete 512-variable integer quartic

Let $R_0, \ldots, R_{309}$ be the literal rows in `direct/equation512.json`, with variables $x_0, \ldots, x_{511}$ in the recorded order. Define

$$\boxed{P_{512}(x_0, \ldots, x_{511}) = \sum_{j=0}^{309} R_j(x_0, \ldots, x_{511})^2.} \tag{11}$$

This defines one fixed integer polynomial, not a family indexed by a program or a theorem predicate. The complete collected coefficient list is `direct/polynomial512.json`.

The parent is already in translated coordinates. Its semantic horizon is $t = 15 + x_0$, not $16 + x_0$; the old nonnegative horizon slack is $x_0 - 1$. The new horizon row is therefore $x_0 - 1 - u^2 = 0$, which is precisely (6). The new rows before removing unused coordinates are recorded verbatim in `direct/specialization_certificate.json`. The exact old coordinate indices removed are

$$329,\ 330,\ 343,\ 344,\ 345.$$

Every other row is unchanged apart from the final order-preserving renumbering. No coordinate whose value is used elsewhere is deleted.

Theorem 5.1. *The polynomial P_{512} has exact degree four, 6,742 nonzero collected monomials, coefficient height 3,904,268, and exactly 512 occurring variables. ZFC proves*

$$[\forall \mathbf{x} \in \mathbb{Z}^{512}\ P_{512}(\mathbf{x}) \neq 0] \longleftrightarrow \mathrm{Con(ZFC)}.$$

Proof. If $P_{512} = 0$, each integer square in (11) is zero. The new horizon row implies the old horizon inequality, the strengthened factorial row implies $F \geq 1$, and (10) gives $r_0 = V^2 + 2F - 2 \geq 0$ without knowing the meaning of V. By (7), the deleted private nonnegativity witnesses can be restored. The unchanged rows and the restored witnesses give a zero of the parent 517-variable system. The retained parent soundness theorem and (2) yield a finite ZFC refutation.

Conversely, translate a finite ZFC refutation into a finite C_{222} derivation using the inherited calibration. Expand it to a finite proof tree. Choose the cofinal horizon (6) beyond all finite bounds, then set $F = t!$ and $V = (Ft^2+1)^t$. The preceding section constructs the deterministic admissible modulus y and the first CRT code. The parent's remaining constructive completeness proof supplies the second CRT data and all shared Pell witnesses. Since $F - 1 \geq 0$, the strengthened factorial guard has three-square witnesses. These values satisfy every new row. Exact parent projection and translation then give a zero of P_{512}.

The coefficient and row data establish the stated finite metrics by exact expansion. Degree four also follows intrinsically: a nonzero homogeneous quadratic part occurs among the residuals, and the sum of their squares cannot vanish identically over $\mathbb{R}$. The quantified transformations use finite syntax, induction, integer arithmetic, CRT, the three-square theorem, and the retained Pell/calibration results, all formalizable in ZFC. Negating the zero-existence equivalence gives the displayed statement. □

This is an equivalence of existence. An arbitrary old zero with a nonsquare horizon slack need not be a new zero. The converse rebuilds a certificate from the represented finite refutation. Confusing these two kinds of reduction would invalidate the completeness argument.

6 A signed factorial output: the 508-variable refinement

The preceding 512-variable equation is a useful control because it changes only three rows of the final parent. A further change at the arithmetic source recovers the factorial output's sign from the equations. Its three private sign witnesses can therefore be removed. Sharing the existing square also permits one additional exact projection in the rebuilt system.

6.1 The changed source

Keep F as an unrestricted integer, retain the cofinal horizon $t = 16 + u^2$, and change the arithmetic definitions to

$$\rho = F^2t^2, \qquad R = 2 + t + x + y + F^2, \qquad K = 2R, \qquad \boxed{y = V(V-1).} \tag{12}$$

The power request remains $V = (\rho+1)^t$, but its meaning is established only after the initial sign argument. The existing source already has the wire for F^2 because its reciprocal product uses $B = \binom{F^2+t}{t}$. Thus (12) adds no fresh square coordinate. The formula for y is quadratic in the existing V, y.

Deleting the factorial guard is not justified by silently assuming $F = t!$. Indeed, before the power requests are interpreted, F and V may have either sign. The crucial simplification is that every integer v satisfies $v(v-1) \geq 0$: either $v \geq 1$ or $v \leq 0$. Consequently the literal new modulus row gives

$$y = V(V-1) \geq 0, \qquad y + 1 \geq 1 \tag{13}$$

before any indexed-Pell or power interpretation. This use of discreteness does not hold for arbitrary real V; the domain here is $\mathbb{Z}$. All independent seed indices are therefore positive, in particular

$$t,\ K,\ F^2t^2 + t,\ F^2 + t,\ y + 1,\ x + 1\ > 0. \tag{14}$$

The remaining seeds are the retained positive encoded representatives. The local constructor bound is also positive on those representatives. Here $x \geq 0$ and every unchanged seed guard are retained. In particular $R \geq 2 + t > 0$ after (14).

The unchanged common budget and guarded frame norm give the same parameter bounds as in the parent. Every seed inequality implies a positive Pell ordinate. Their shared auxiliary product is nonzero, so its norm establishes the previously required auxiliary-coordinate bound. The signed indexed-Pell theorem is therefore licensed in the same manner as in the parent. The old positivity certificate is replaced by these actual seed formulas; it is not cited unchanged after the deletion of its factorial premise. No exceptional zero-output case or parity restriction on the horizon is needed.

6.2 Recovering powers and the factorial in the correct order

Let D denote the proposed factorial binomial output. The retained natural remainders give $D \geq 1$ and a common radix $q \geq 2$. The request q^t orients the shared t-indexed Pell pair. Thus

$$V = (F^2t^2 + 1)^t \geq 1.$$

The raw ceiling rows give $DF = V + s$, where $s \geq 0$. Since $D \geq 1$, this forces $F \geq 1$ before any factorial value is assumed. Since $t \geq 16$, $V \geq F^2t^2 + 1 \geq F + 2$, and therefore

$$y - F - 1 = V(V - 1) - F - 1 \geq 0.$$

The old endpoint slack can now be recovered.

The reciprocal raw index row is $C + F^2 + 1 = Bi_q$, with $C \geq 1$ and $B \geq 0$ still guarded. It gives $B > 0$. In the signed power request $N = B^K$, the only possible inverse case is $B = 1, N = -1$. But then the other base at exponent K is $N + V + 1 \geq 2$. That request orients the K-indexed pair and excludes the inverse case. The common radix has its intended value

$$q = N(N + V + 1)^K > V^2.$$

Every remaining seed is similarly oriented by a request at base q or $q + 1$. This establishes all power meanings without first interpreting the factorial or the binomials whose digits are under validation.

Now $\rho = F^2t^2 \geq t$, and for $0 \leq j \leq t$,

$$\binom{\rho + t}{j} \leq \binom{\rho + t}{t} \leq (\rho + 1)^t = V < q.$$

The first digit graph therefore really gives $D = \binom{\rho+t}{t}$. The exact factorial argument is recorded next, rather than inferred from finite sample calculations.

Lemma 6.1 (Self-bounding factorial at the squared radius)**.** *Let F, t be positive integers, $\rho = F^2t^2$, $V = (\rho + 1)^t$, and $D = \binom{\rho+t}{t}$. Then*

$$F - 1 < V/D \leq F \quad \Longleftrightarrow \quad F = t!.$$

Proof. Write $G = t!$. The exact identity is

$$\frac{V}{D} = G \prod_{i=1}^{t} \left(1 - \frac{i - 1}{\rho + i}\right).$$

All factors lie in $(0, 1]$, and

$$\sum_{i=1}^{t} \frac{i - 1}{\rho + i} < \frac{1}{2F^2}.$$

The elementary product bound gives $G(1-1/(2F^2)) < V/D \le G$. If $F-1 < V/D \le F$, then $G > F-1$, so $G \ge F$. On the other hand,

$$G < \frac{F}{1-1/(2F^2)} = F + \frac{F}{2F^2-1} \le F+1,$$

and integrality gives $G \le F$. Thus $F = G$. Conversely, if $F = G$, then $F-1 < F-1/(2F) < V/D \le F$; when $F = 1$ the same strict inequality also holds. The ceiling remainder $DF - V$ lies in $[0, D-1]$, giving the required natural ceiling witnesses. □

The source ceiling rows are precisely the inequality in this lemma, so $F = t!$. The second digit is also safe: for $0 \le j \le t$,

$$\binom{F^2+t}{j} \le (F^2+t)^t \le (\rho+1)^{2t} = V^2 < q.$$

It recovers the reciprocal product B and $B \ge 2$. The later amplified CRT and derived-label bounds are preserved: after $F \ge 1$ the new $R = 2+t+x+y+F^2$ is at least the old $2+t+x+y+F$. The same local majorant exponent, factorial growth floor, representative bounds, and strengthened CRT precision therefore apply. No preceding bootstrap step used that later factorial growth estimate.

6.3 Completeness and the literal polynomial

Choose the cofinal padded horizon as before and set $F = t!$, $\rho = F^2t^2$, and $V = (\rho+1)^t$. A slightly stronger pending-product bound is useful. Put $a = Ft \ge 16$, $L = a+1$, and $B_0 = a^2+1$. There are at most $t+1$ pending weights, each at most L, so $S_i \le L^{t+1}$. Since $a \ge 3$ and $t \ge 2$,

$$B_0 > (a-1)L, \qquad (a-1)^t \ge (a-1)^2 \ge L.$$

It follows, using integer powers only, that

$$V = B_0^t > L^t(a-1)^t \ge L^{t+1} \ge S_i.$$

Now $V \ge 3$, so the prescribed modulus $y = V(V-1)$ is at least V and exceeds every accumulator. Also $F \mid (V-1)$ because $V \equiv 1 \pmod F$, and hence $F \mid y$. We have already established $y \ge F+1$. These are exactly the inherited CRT requirements; the earlier sufficient convenience $y > V^2$ is unnecessary. The first CRT is valid. The larger new R satisfies every subsequent lower bound. The factorial lemma supplies the ceiling witnesses, the unchanged digit arguments supply their quotients, and simultaneous Pell completeness supplies the common frame. Thus every finite refutation gives a zero of the new arithmetic system.

The source is then factored, projected through monic integral definitions, and translated by the same recorded pipeline as its matched parent. The principal result is

$$\boxed{P_{508}(x_0,\dots,x_{507}) = \sum_{j=0}^{307} Q_j(x_0,\dots,x_{507})^2 = 0.} \tag{15}$$

The full Q_j are printed in Appendix A and supplied in `direct/principal/equation.json`. Their full collected expansion is `direct/principal/expanded_polynomial.json`.

Theorem 6.2 (Principal unrestricted-integer equation)**.** *The polynomial in (15) has 508 occurring integer variables, 308 quadratic residuals, exact degree four, 6,545 collected monomials, and coefficient height 996,521. ZFC proves*

$$\left[\forall \mathbf{x} \in \mathbb{Z}^{508}\ P_{508}(\mathbf{x}) \ne 0\right] \longleftrightarrow \mathrm{Con(ZFC)}.$$

Proof. The product of consecutive integers establishes the modulus sign, and hence all seed signs, without assuming a factorial or power meaning. The ordered recovery gives the same bounded local proof-certificate semantics, whose soundness gives (2). The constructive paragraph above proves the converse from any finite refutation. Every source modification is identified by an exact quadratic-row certificate. The subsequent monic eliminations and integer translations have recorded integral inverse maps, so they preserve integer-zero existence. Finally a sum of integral squares vanishes exactly when all its residuals vanish. Exact coefficient expansion establishes the finite metrics, including a nonzero degree-four part. All quantified mathematical steps are formalizable in ZFC, yielding (1) for $D = \mathbb{Z}$ and $n = 508$. □

The matched parent has 517 variables, 310 rows, 6,774 terms, and coefficient height 3,904,267. This refinement improves all four of those quantities while retaining degree four and unrestricted integer unknowns. It removes about 1.74% of the variables and 3.38% of the monomials; coefficient height falls by about 74.48%. Those are measured improvements in this particular direct construction, not a claim that 508 is the smallest possible integer-variable count.

7 Ten positive unknowns with an 89-operation evaluator

The direct result above keeps modest coefficients and unrestricted integer unknowns. A complementary calculation shortens the existing compact native route. This section's domain is explicitly $\mathbb{Z}_{>0}^{10}$, and its large fixed coefficients are inherited from the coherent compact44 BFS tuple. It is not a 10-variable unrestricted-integer result.

Polynomial	Variables	Operations	Exact degree
Uncentered	10	89	288
Upper center	10	91	283
Sharp center	10	92	278
Direct, uncentered	10	94	262
Direct, upper center	10	96	258
Direct, sharp center	10	97	254

The centered parents in the incoming saturated-Pell packet cost, in the same order, 92/283, 93/278, 98/258, and 99/254, where each pair is operations/degree [3]. Thus all four centered companions improve cost at unchanged degree. The uncentered 89-operation equation is a new cost/degree tradeoff. The earlier square-gap and auxiliary-index ideas in the incoming structural-compression packet are explicitly credited [4]; the new step is their combination with coprime Pell fusion and the resulting replacement of the transport majorant.

7.1 Complete polynomial definition

Use the positive unknowns

$$J, F, \alpha, i, T, w, Y, n, \eta, \zeta.$$

On the direct chart replace w by a positive unknown x. The four constants β, K_0, M_C, M_F are the entire compact44 BFS tuple supplied in `positive/coefficients/`. In particular $\beta = 2^{590069} - 1$. The full fixed finite recipe, including its 436-character seed and all finite index bounds, is `positive/coefficients/recipe_compact44.py`. It reconstructs every coefficient and checks its complete raw-byte hash. The high mask M_F is already shifted; no second shift or mixture with the differently ordered original44 tuple is permitted.

Define only polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & \mu &= q^2 + 1, & & \\
k &= \eta + \zeta, & c &= kY + \eta, & & \\
X &= wq \quad \text{(ordinary)}, & X &= q + x \quad \text{(direct)}, & & \\
A &= \mu Y(X+1) + 1, & H &= 4A - 5, & \Delta &= A^2 - 1, \\
V_0 &= X(\mu Y)^2, & E &= V_0(V_0+1)k^2 + 1, & & \\
R &= [(q-1)(q-F) + \alpha](q^2-1) & & & & \\
&\quad + (M_C + qM_F)J, & & & & \\
e &= k - R - 1, & C &= q - F - \alpha, & U &= ic^2, \\
D_f &= 1 + \Delta U^2 E, & Q &= \Delta^2 U^2 E, & \delta &= Q - 1, \\
b &= c(T+\alpha) - R, & V &= D_f b - c, & D_y &= QV^2 - 1, \\
K_+ &= X(3c - 2X) + 2. & & & &
\end{aligned}$$

The ordinary chart uses

$$B = (K_0 + w)C - F, \quad h = q - 1, \quad P = q^3(q-1), \quad D_{\text{low}} = 0,$$

and the direct chart uses

$$B = (qK_0 + X)C - qF, \quad h = q(q-1), \quad P = q^4(q-1), \quad D_{\text{low}} = (q-1)X.$$

Select a center

$$C_g = 0, \qquad C_g = (A+1)UE, \qquad \text{or} \qquad C_g = AUE,$$

respectively. Finally set

$$\begin{aligned}
M &= P\mu^2 H, \\
Z &= P\mu^2\{H(n + V + C_g - bB) + K_+\} \\
&\quad + H\{\mu^2(hY - q^3 B - D_{\text{low}}) - Pe\}, \\
s &= \delta Z^2, \qquad u = \delta M^2 E D_f, \qquad v = M^2 D_y.
\end{aligned}$$

The single equation, for any of the six choices, is

$$\boxed{N = (s - u - v)^2 - 4uv = 0.} \tag{16}$$

All abbreviations use only addition, subtraction, multiplication, and fixed nonnegative integer exponents. The literal JSON circuits expand every one of them; no square root or auxiliary predicate occurs in N.

7.2 The auxiliary coordinate is a sufficient transport bound

The inherited coprime recovery, before any transport sign conclusion, gives positive integers τ, f, y with

$$E = \tau^2, \quad D_f = f^2, \quad I = U\tau, \quad Q = (\Delta I)^2, \quad QV^2 - (Q-1)y^2 = 1. \tag{17}$$

For clarity, here I is a proof-only recovered integer, not an additional unknown. The coprime step is valid because $D_f \equiv 1 \pmod{E}$. The two rational radicands ED_f and D_y/δ cannot coincide: after multiplying by δ and reducing modulo E, equality would give $0 \equiv -1$. Thus the rational signed-sum argument makes

both roots rational; square-class coprimality separates E and D_f, and the retained adjacent-coefficient descent makes y integral. This order does not presuppose the later projection or the sign of B.

Write $\psi_A(m)$ for the Pell ordinate at base A. The retained uniform folding and first-index arguments give $c = \psi_A(p)$ for an odd positive p, together with $c > 8q^5$, $c > 2X$, $H > 3X$, and $K_+ > 0$. The native mask bound $R < q^4 + (\alpha+2)q^2 < c\alpha/4$ then gives

$$b = c(T+\alpha) - R > c.$$

Since $c \geq 3$, the odd index p is at least three, so $c \geq 4A^2 - 1 = 4\Delta + 3$.

Lemma 7.1 (Fifth-index gap after fusion)**.** *Under the recovered conditions above,*

$$b > 8Q, \qquad g := \tau f < Q, \qquad 0 < y - V < b/4.$$

Proof. From $b > c$,

$$V = D_f b - c > (D_f - 1)c = \Delta I^2 c > 4Q.$$

Put $S = \Delta I$, so $Q = S^2$. Pell classification applied to $(SV)^2 - (S^2-1)y^2 = 1$ gives $SV = \chi_S(\nu)$ with positive odd ν. The values at $\nu = 1, 3$ give $V = 1, 4Q - 3$, both excluded. Thus $\nu \geq 5$ and

$$V \geq 16Q^2 - 20Q + 5 > 8Q^2.$$

Since $D_f < Q$, this yields $b = (V + c)/D_f > 8Q$. Also $f < AI$, so $g < AUE < Q$. Finally $Q - 1 > 2D_f$ and

$$0 < y - V = \frac{V^2 - 1}{(Q-1)(y+V)} < \frac{V}{2(Q-1)} < \frac{b}{4}.$$

Each inequality precedes the transport sign recovery. □

For explanation only, divide the carrier by M and put $W = 1 + hb$. The exact identity is

$$\frac{Z}{M} = n + V + C_g + \frac{K_+}{H} + \frac{Y}{q^3} - \frac{WB}{h} - \frac{e}{\mu^2} - d, \tag{18}$$

where $d = 0$ ordinarily and $d = X/q^4$ directly. At a norm zero, $Z/M = \varepsilon_g g + \varepsilon_y y$. Since $Q > X$, we have $g + d < 2Q < b/4$. Also $b - e = c(T+\alpha) - k + 1 > 0$, so $e/\mu^2 < b/4$ even when e is negative. If $B \leq -1$, solving (18) for n gives

$$n < \frac{3b}{4} + \frac{WB}{h} \leq \frac{3b}{4} - b - \frac{1}{h} < 0.$$

This contradicts the domain of n. A positive center C_g only decreases the right side. Thus all six equations force $B \geq 0$ using the existing coordinate b itself; the old separately computed transport majorant is unnecessary. The cheapest pair can also discard the center.

7.3 Completeness and the consistency theorem

After $B \geq 0$ is established, the unchanged source arguments recover $C > 0$, $3q < R < q^4 - 2$, the exact index $p = R$, denominator divisibilities, and $X = 2^R$. They reconstruct the same unscaled native source witness; the scaled strong-Pell witness is not divided and reused without proof. The retained native/controller/refutation calibration therefore applies.

Conversely, start with the parent's canonical data supplied by a finite refutation. Choose a fresh strong Pell period with $c^2\tau \mid I$, which is possible by the finite order of the determinant-one Pell matrix modulo

$c^2\tau$. Put $i = I/(c^2\tau) > 0$. The retained auxiliary-completeness construction gives positive T, the auxiliary value V, and a gap $u_* = y - V > 4I^2$. It also gives

$$\kappa = K_+/H < c < I, \quad s_Y = Y/q^3 < Y < c < I, \quad UE = I\tau < I^2.$$

First work on the ordinary chart, with $B_t = (K_0 + w)C - F$ and $W = 1 + (q-1)b$. The inherited canonical divisibilities make κ, s_Y, $t_* = B_t/(q-1) \geq 0$, and $r_* = e/\mu^2 \geq 0$ integers. Choose

$$n = g - C_g + u_* - \kappa - s_Y + Wt_* + r_*.$$

For the uncentered choice $g > 2c > \kappa + s_Y$ proves $n > 0$. For the centered choices, $0 < C_g - g < 2UE$ and the total negative loss is less than $2I^2 + 2I < 4I^2 < u_*$. The norm then vanishes. The direct chart takes $x = X - q > 0$ and adds the positive integral correction $(W_{\text{direct}} - W_{\text{ordinary}})t_* + X/q^4$ to n. Here the inherited canonical power divisibility gives $q^4 \mid X$. Thus every finite refutation supplies positive zeros in both charts.

It follows that, for every polynomial N in the table,

$$\mathrm{ZFC} \vdash [\forall \mathbf{z} \in \mathbb{Z}_{>0}^{10}\ N(\mathbf{z}) \neq 0] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

The complete new proof, with the precise inherited theorem interfaces, is `positive/PROOF.md`.

7.4 Exact degree, rather than a syntactic upper estimate

Let $q_0 = \beta J$, $k_0 = \eta + \zeta$, and $x_0 = q_0 + x$. Uniform formal scaling of all ten variables gives the following nonzero highest homogeneous parts for the uncentered pair:

$$N_{288,\text{top}} = -2^{10} q_0^{152} w^{30} Y^{62} i^{10} k_0^{32} (q_0 - F)^2,$$
$$N_{262,\text{top}} = -2^{10} q_0^{126} x_0^{30} Y^{62} i^{10} k_0^{32} (q_0 - F)^2.$$

They are nonzero for the actual positive fixed β. In the formal Laurent-series field, the rational remainder in $Z/M = V + C_g + \mathcal{E}$ has degree at most six. The centered radical defects have degrees at least seventeen, so the changed transport cannot alter their leading terms. Pairing the two signs of each radical yields the exact centered degrees in the table. The proof provides the complete leading forms and global degree bound; exact one-variable specialization checks are supplementary transcription controls, not the sole degree proof.

8 Verification, reproducibility, and evidence boundaries

The release contains a portable `verify_all.py`. It checks the exact finite artifacts without querying a service or fetching a repository. Python 3.10 or later and the standard library suffice for the supported checkers. The complete equation specifications are the literal JSON files, not the human-readable metric summaries.

The principal direct checker independently compares the modified source against its frozen parent, checks the complete square budget and the new seed/sign prerequisite rows, and replays the factorization, monic projection, and translation maps. It expands every quadratic residual square and compares every collected integer coefficient with the supplied expansion. It does not import the equation builder to decide whether the final coefficients are correct. The 512-variable checker separately checks all three changed rows and the exact order-preserving coordinate deletion.

For the positive companions, the independent verifier compares all six complete parent-to-child graphs, reconstructs all four actual fixed coefficients, checks every live operation, expands the universal two-root

norm identity, and checks the cleared carrier identity. Exact signed evaluations, evaluations with the actual fixed coefficients, and univariate polynomial degree replays supplement those exact structural identities. Thirty deliberate circuit or binding corruptions are rejected. The complete homogeneous-part argument in Section 7, rather than the degree of one specialization alone, supplies the global degree proof.

Principal direct finite certificate	Checked count
Changed or deleted source rows	5
Recognized affine square roots / Horn product links	17 / 11
Integral coordinate restorations	222
Final integer translations	13
Distinct atoms in the common square budget	41
Independent positive seed indices	16
Collected coefficients in the final quartic	6,545

The supporting semantic regression program checks small factorial candidates, concrete encoding moduli and accumulators, signed products of consecutive integers, and the strengthened bound $S_i < V$. It also retains a negative control showing why arbitrary admissible witnesses cannot be restricted to squares. Exact test counts are recorded in its receipts. These tests illustrate the formulas and detect transcription errors; none is used as a proof of an assertion quantified over all integers.

Supported checkers were run both normally and with Python optimization enabled, so their correctness checks do not rely on removable `assert` statements. The assembled release was checked from its own directory. Deterministic regeneration of the principal direct artifacts is separately recorded. The residual appendix is generated from the complete principal equation and checked byte-for-byte by `report/generate_residuals.py--check`.

For integration, copy this packet as a new research contribution. Preserve the supplied source ledgers and the distinction between the new lemmas and the inherited calibration. The main article's editable source is `report/main.tex`; compile it with a standard TeX Live installation. The release manifest authenticates the supplied files. Rebuilding receipts or the PDF can change their bytes; verify the original manifest before rebuilding if an unchanged-archive check is desired.

9 Further research questions

Which other witnesses can be chosen canonically? The useful criterion is constructive completeness with an explicit order of choices. Cofinality by itself is insufficient in the presence of residue conditions. A systematic pass over the remaining free bounds could identify parameters that can be replaced by existing powers or monotone polynomial families without introducing fresh coordinates.

Can the number of independent exponent certificates be reduced? The direct construction shares a Pell frame but retains multiple certified indices. An identity between genuine requested powers can eliminate a certificate only if its signs, range bounds, and noncircular recovery order remain justified. The complete literal source is small enough for such changes to be measured rather than estimated.

Can proof-tree shape improve the accumulator bound? The proof above uses the coarse bound $t + 1$ on the number of pending formulas. Depth-first expansion bounds it more sharply in terms of proof-tree depth. Turning that observation into a smaller polynomial requires encoding or uniformly controlling that depth without spending more unknowns than are saved. It may be more useful for exponent size than for variable count.

Can the compact positive equations lower both degree and circuit cost? The positive companions exchange transport and norm-centering costs for degree. Their constants, source calibration, and domains

should remain fixed while comparing transformations. A low gate count with very large fixed coefficients should not be interpreted as a comparable bit-complexity result.

What is the best formalization target? The new cofinal selection and modulus theorems are short and separable from the inherited compiler. Formalizing these lemmas and the exact finite coefficient checker would provide a useful next verification layer. A fully internal ZFC proof object or an end-to-end Lean/Rocq development remains a separate task from the ordinary proof delivered here.

How should simplicity be scored? A reproducible comparison should retain the vector

$$(\text{domain},\ \text{variables},\ \text{degree},\ \text{height},\ \text{monomials},\ \text{live operations})$$

and report which entries improve. The constructions here do not establish a global lower bound in any of these measures.

References

[1] Vladimir Reshetnikov, *ProveIt: Diophantine consistency research*, snapshot `45b399ec51dfbecb891dd51d8a9e4b6b7c361ac6`, 24 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/45b399ec51dfbecb891dd51d8a9e4b6b7c361ac6/Computability/DiophantineConsistency.

[2] *Diophantine consistency equations with powered saturation and factorial carriers*, Round 14, 24 October 2026. Incoming archive `ConZFC_Powered_Saturation_Factorial_Carriers_Round14_2026-10-24.zip` at the snapshot in [1]. The exact ancestor files and hashes are retained in this packet's provenance records.

[3] *ConZFC saturated Pell*, 24 October 2026. Incoming archive `ConZFC_Saturated_Pell_2026-10-24.zip` at the same snapshot. Its complete source coefficient binding and applicable proof documents are retained with the positive companions.

[4] *Removing Redundancy from Diophantine Consistency Equations*, 24 October 2026. Incoming archive `con-zfc-structural-compression-2026-10-24.zip`, same snapshot.

[5] Jonas Bayer, Marco David, Malte Haßler, Yuri Matiyasevich, and Dierk Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v3. https://arxiv.org/html/2506.20909v3. Used for context and the standard three-square representation, not as a substituted universal-polynomial parameter in the delivered direct equation.

A Complete quadratic residuals of the principal equation

The unknowns are $x_0, \dots, x_{507} \in \mathbb{Z}$, with the zero-based coordinate order of `direct/principal/equation.json`. The following list contains every term of all 308 integral quadratic residuals. The single principal equation is exactly

$$P_{508}(x_0, \dots, x_{507}) = \sum_{j=0}^{307} Q_j(x_0, \dots, x_{507})^2 = 0.$$

The symbols Q_j are polynomial abbreviations, not additional unknowns. Every coefficient is printed below; no term is omitted.

$$\begin{aligned}
Q_0 = {} & -225 - 30\,x_0 - x_0^2 + x_4 \\
Q_1 = {} & -15 - x_0 + x_0\,x_4 + 15\,x_4 \\
& - x_4\,x_{190} + x_4\,x_{193} + x_{177} - x_{180} \\
Q_2 = {} & -x_0\,x_2 - 16\,x_2 + x_9 \\
Q_3 = {} & -1 + x_1 - x_2\,x_7 - x_3 \\
& - x_7 \\
Q_4 = {} & -1 + x_1 - x_8 - x_8\,x_9 \\
Q_5 = {} & -15 - x_0 - x_3^2 + x_{190} \\
& - x_{193} \\
Q_6 = {} & -14 - x_0 + x_{18} - x_{19}\,x_{21} \\
& + x_{190} - x_{193} \\
Q_7 = {} & -x_{10}^2 - 2\,x_{10}\,x_{11} - x_{11}^2 + x_{22} \\
Q_8 = {} & -71 - 102\,x_{10} - 36\,x_{10}^2 - 96\,x_{10}\,x_{11} \\
& - 24\,x_{10}\,x_{22} - 136\,x_{11} - 64\,x_{11}^2 - 32\,x_{11}\,x_{22} \\
& - 34\,x_{22} - 4\,x_{22}^2 + x_{23} \\
Q_9 = {} & -x_{10}^2 - 4\,x_{10}\,x_{23} - 4\,x_{23}^2 + x_{24} \\
Q_{10} = {} & -55 - 90\,x_{10} - 36\,x_{10}^2 - 192\,x_{10}\,x_{23} \\
& - 24\,x_{10}\,x_{24} - 240\,x_{23} - 256\,x_{23}^2 - 64\,x_{23}\,x_{24} \\
& - 30\,x_{24} - 4\,x_{24}^2 + x_{25} \\
Q_{11} = {} & -x_{11}^2 - 2\,x_{11}\,x_{12} - x_{12}^2 + x_{26} \\
Q_{12} = {} & -3\,x_{10} - x_{10}^2 - 4\,x_{10}\,x_{11} - 4\,x_{10}\,x_{26} \\
& - 8\,x_{11} - 4\,x_{11}^2 - 8\,x_{11}\,x_{26} - 8\,x_{26} \\
& - 4\,x_{26}^2 + x_{27} \\
Q_{13} = {} & -x_{10}^2 - 2\,x_{10}\,x_{12} - x_{12}^2 + x_{28} \\
Q_{14} = {} & -150 - 294\,x_{10} - 144\,x_{10}^2 - 192\,x_{10}\,x_{11} \\
& - 192\,x_{10}\,x_{12} - 48\,x_{10}\,x_{22} - 48\,x_{10}\,x_{28} - 200\,x_{11} \\
& - 64\,x_{11}^2 - 128\,x_{11}\,x_{12} - 32\,x_{11}\,x_{22} - 32\,x_{11}\,x_{28} \\
& - 192\,x_{12} - 64\,x_{12}^2 - 32\,x_{12}\,x_{22} - 32\,x_{12}\,x_{28} \\
& - 50\,x_{22} - 4\,x_{22}^2 - 8\,x_{22}\,x_{28} - 48\,x_{28} \\
& - 4\,x_{28}^2 + x_{29} \\
Q_{15} = {} & -36 - 24\,x_{27} - 4\,x_{27}^2 - 8\,x_{27}\,x_{29} \\
& - 24\,x_{29} - 4\,x_{29}^2 + x_{30}
\end{aligned}$$

$$
\begin{aligned}
Q_{16} = {} & -4 - 16\,x_{12} - 16\,x_{12}^2 - 16\,x_{12}\,x_{26} \\
& - 16\,x_{12}\,x_{28} - 8\,x_{26} - 4\,x_{26}^2 - 8\,x_{26}\,x_{28} \\
& - 8\,x_{28} - 4\,x_{28}^2 + x_{31} \\
Q_{17} = {} & -88 - 114\,x_{10} - 36\,x_{10}^2 - 96\,x_{10}\,x_{11} \\
& - 48\,x_{10}\,x_{12} - 24\,x_{10}\,x_{22} - 48\,x_{10}\,x_{28} - 24\,x_{10}\,x_{31} \\
& - 152\,x_{11} - 64\,x_{11}^2 - 64\,x_{11}\,x_{12} - 32\,x_{11}\,x_{22} \\
& - 64\,x_{11}\,x_{28} - 32\,x_{11}\,x_{31} - 72\,x_{12} - 16\,x_{12}^2 \\
& - 16\,x_{12}\,x_{22} - 32\,x_{12}\,x_{28} - 16\,x_{12}\,x_{31} - 38\,x_{22} \\
& - 4\,x_{22}^2 - 16\,x_{22}\,x_{28} - 8\,x_{22}\,x_{31} - 72\,x_{28} \\
& - 16\,x_{28}^2 - 16\,x_{28}\,x_{31} - 36\,x_{31} - 4\,x_{31}^2 \\
& + x_{32} \\
Q_{18} = {} & -60 - 15\,x_{10} - x_{10}^2 - 12\,x_{10}\,x_{11} \\
& - 16\,x_{10}\,x_{12} - 4\,x_{10}\,x_{26} - 96\,x_{11} - 36\,x_{11}^2 \\
& - 96\,x_{11}\,x_{12} - 24\,x_{11}\,x_{26} - 128\,x_{12} - 64\,x_{12}^2 \\
& - 32\,x_{12}\,x_{26} - 32\,x_{26} - 4\,x_{26}^2 + x_{33} \\
Q_{19} = {} & -60 - 96\,x_{10} - 36\,x_{10}^2 - 12\,x_{10}\,x_{11} \\
& - 96\,x_{10}\,x_{12} - 24\,x_{10}\,x_{28} - 15\,x_{11} - x_{11}^2 \\
& - 16\,x_{11}\,x_{12} - 4\,x_{11}\,x_{28} - 128\,x_{12} - 64\,x_{12}^2 \\
& - 32\,x_{12}\,x_{28} - 32\,x_{28} - 4\,x_{28}^2 + x_{34} \\
Q_{20} = {} & -144 - 48\,x_{33} - 4\,x_{33}^2 - 8\,x_{33}\,x_{34} \\
& - 48\,x_{34} - 4\,x_{34}^2 + x_{35} \\
Q_{21} = {} & -4 - 8\,x_{10} - 4\,x_{10}^2 - 8\,x_{10}\,x_{12} \\
& - 8\,x_{10}\,x_{22} - 8\,x_{10}\,x_{26} - 8\,x_{12} - 4\,x_{12}^2 \\
& - 8\,x_{12}\,x_{22} - 8\,x_{12}\,x_{26} - 8\,x_{22} - 4\,x_{22}^2 \\
& - 8\,x_{22}\,x_{26} - 8\,x_{26} - 4\,x_{26}^2 + x_{36} \\
Q_{22} = {} & -88 - 152\,x_{10} - 64\,x_{10}^2 - 160\,x_{10}\,x_{12} \\
& - 64\,x_{10}\,x_{26} - 32\,x_{10}\,x_{28} - 32\,x_{10}\,x_{36} - 186\,x_{12} \\
& - 100\,x_{12}^2 - 80\,x_{12}\,x_{26} - 40\,x_{12}\,x_{28} - 40\,x_{12}\,x_{36} \\
& - 72\,x_{26} - 16\,x_{26}^2 - 16\,x_{26}\,x_{28} - 16\,x_{26}\,x_{36} \\
& - 38\,x_{28} - 4\,x_{28}^2 - 8\,x_{28}\,x_{36} - 36\,x_{36} \\
& - 4\,x_{36}^2 + x_{37} \\
Q_{23} = {} & -55 - 90\,x_{10} - 36\,x_{10}^2 - 96\,x_{10}\,x_{11} \\
& - 24\,x_{10}\,x_{22} - 120\,x_{11} - 64\,x_{11}^2 - 32\,x_{11}\,x_{22} \\
& - 30\,x_{22} - 4\,x_{22}^2 + x_{38} \\
Q_{24} = {} & -x_{10}^2 - 4\,x_{10}\,x_{38} - 4\,x_{38}^2 + x_{39} \\
Q_{25} = {} & -55 - 90\,x_{10} - 36\,x_{10}^2 - 192\,x_{10}\,x_{38} \\
& - 24\,x_{10}\,x_{39} - 240\,x_{38} - 256\,x_{38}^2 - 64\,x_{38}\,x_{39} \\
& - 30\,x_{39} - 4\,x_{39}^2 + x_{40} \\
Q_{26} = {} & -x_{11}^2 - 4\,x_{11}\,x_{40} - 4\,x_{40}^2 + x_{41} \\
Q_{27} = {} & -95 - 158\,x_{10} - 64\,x_{10}^2 - 128\,x_{10}\,x_{11} \\
& - 64\,x_{10}\,x_{22} - 160\,x_{11} - 64\,x_{11}^2 - 64\,x_{11}\,x_{22} \\
& - 78\,x_{22} - 16\,x_{22}^2 + x_{42}
\end{aligned}
$$

$$
\begin{aligned}
Q_{28} = {} & -176 - 216\,x_{11} - 64\,x_{11}^2 - 96\,x_{11}\,x_{12} \\
& - 32\,x_{11}\,x_{26} - 32\,x_{11}\,x_{42} - 162\,x_{12} - 36\,x_{12}^2 \\
& - 24\,x_{12}\,x_{26} - 24\,x_{12}\,x_{42} - 54\,x_{26} - 4\,x_{26}^2 \\
& - 8\,x_{26}\,x_{42} - 52\,x_{42} - 4\,x_{42}^2 + x_{43} \\
Q_{29} = {} & -56 - 120\,x_{10} - 64\,x_{10}^2 - 96\,x_{10}\,x_{12} \\
& - 32\,x_{10}\,x_{28} - 32\,x_{10}\,x_{43} - 90\,x_{12} - 36\,x_{12}^2 \\
& - 24\,x_{12}\,x_{28} - 24\,x_{12}\,x_{43} - 30\,x_{28} - 4\,x_{28}^2 \\
& - 8\,x_{28}\,x_{43} - 28\,x_{43} - 4\,x_{43}^2 + x_{44} \\
Q_{30} = {} & -x_{12}^2 - 2\,x_{12}\,x_{13} - x_{13}^2 + x_{45} \\
Q_{31} = {} & -x_{10}^2 - 4\,x_{10}\,x_{12} - 4\,x_{10}\,x_{45} - 4\,x_{12}^2 \\
& - 8\,x_{12}\,x_{45} - 4\,x_{45}^2 + x_{46} \\
Q_{32} = {} & -4\,x_{10}^2 - 8\,x_{10}\,x_{11} - 8\,x_{10}\,x_{26} - 8\,x_{10}\,x_{28} \\
& - 4\,x_{11}^2 - 8\,x_{11}\,x_{26} - 8\,x_{11}\,x_{28} - 4\,x_{26}^2 \\
& - 8\,x_{26}\,x_{28} - 4\,x_{28}^2 + x_{47} \\
Q_{33} = {} & -16\,x_{10}^2 - 32\,x_{10}\,x_{12} - 32\,x_{10}\,x_{28} - 32\,x_{10}\,x_{45} \\
& - 16\,x_{10}\,x_{46} - 16\,x_{10}\,x_{47} - 4\,x_{12} - 16\,x_{12}^2 \\
& - 32\,x_{12}\,x_{28} - 32\,x_{12}\,x_{45} - 16\,x_{12}\,x_{46} - 16\,x_{12}\,x_{47} \\
& - 16\,x_{28}^2 - 32\,x_{28}\,x_{45} - 16\,x_{28}\,x_{46} - 16\,x_{28}\,x_{47} \\
& - 4\,x_{45} - 16\,x_{45}^2 - 16\,x_{45}\,x_{46} - 16\,x_{45}\,x_{47} \\
& - 2\,x_{46} - 4\,x_{46}^2 - 8\,x_{46}\,x_{47} - 4\,x_{47}^2 \\
& + x_{48} \\
Q_{34} = {} & -12 - 39\,x_{12} - 25\,x_{12}^2 - 40\,x_{12}\,x_{28} \\
& - 20\,x_{12}\,x_{31} - 32\,x_{28} - 16\,x_{28}^2 - 16\,x_{28}\,x_{31} \\
& - 16\,x_{31} - 4\,x_{31}^2 + x_{49} \\
Q_{35} = {} & -4 - 8\,x_{11} - 4\,x_{11}^2 - 8\,x_{11}\,x_{26} \\
& - 8\,x_{26} - 4\,x_{26}^2 + x_{50} \\
Q_{36} = {} & -4 - 16\,x_{11} - 16\,x_{11}^2 - 8\,x_{11}\,x_{12} \\
& - 32\,x_{11}\,x_{26} - 16\,x_{11}\,x_{50} - 4\,x_{12} - x_{12}^2 \\
& - 8\,x_{12}\,x_{26} - 4\,x_{12}\,x_{50} - 16\,x_{26} - 16\,x_{26}^2 \\
& - 16\,x_{26}\,x_{50} - 8\,x_{50} - 4\,x_{50}^2 + x_{51} \\
Q_{37} = {} & -3\,x_{11} - x_{11}^2 - 4\,x_{11}\,x_{12} - 4\,x_{11}\,x_{51} \\
& - 8\,x_{12} - 4\,x_{12}^2 - 8\,x_{12}\,x_{51} - 8\,x_{51} \\
& - 4\,x_{51}^2 + x_{52} \\
Q_{38} = {} & -49 - 28\,x_{52} - 4\,x_{52}^2 + x_{53} \\
Q_{39} = {} & -144 - 24\,x_{12} - x_{12}^2 - 8\,x_{12}\,x_{52} \\
& - 4\,x_{12}\,x_{53} - 96\,x_{52} - 16\,x_{52}^2 - 16\,x_{52}\,x_{53} \\
& - 48\,x_{53} - 4\,x_{53}^2 + x_{54} \\
Q_{40} = {} & -3\,x_{11} - x_{11}^2 - 4\,x_{11}\,x_{12} - 4\,x_{11}\,x_{54} \\
& - 8\,x_{12} - 4\,x_{12}^2 - 8\,x_{12}\,x_{54} - 8\,x_{54} \\
& - 4\,x_{54}^2 + x_{55} \\
Q_{41} = {} & -144 - 48\,x_{49} - 4\,x_{49}^2 - 8\,x_{49}\,x_{55} \\
& - 48\,x_{55} - 4\,x_{55}^2 + x_{56}
\end{aligned}
$$

$$
\begin{aligned}
Q_{42} = &-192 - 27\,x_{10} - x_{10}^2 - 8\,x_{10}\,x_{49} \\
&- 4\,x_{10}\,x_{56} - 112\,x_{49} - 16\,x_{49}^2 - 16\,x_{49}\,x_{56} \\
&- 56\,x_{56} - 4\,x_{56}^2 + x_{57} \\
Q_{43} = &-49 - 28\,x_{57} - 4\,x_{57}^2 + x_{58} \\
Q_{44} = &-144 - 24\,x_{12} - x_{12}^2 - 8\,x_{12}\,x_{57} \\
&- 4\,x_{12}\,x_{58} - 96\,x_{57} - 16\,x_{57}^2 - 16\,x_{57}\,x_{58} \\
&- 48\,x_{58} - 4\,x_{58}^2 + x_{59} \\
Q_{45} = &-3\,x_{10} - x_{10}^2 - 4\,x_{10}\,x_{12} - 4\,x_{10}\,x_{59} \\
&- 8\,x_{12} - 4\,x_{12}^2 - 8\,x_{12}\,x_{59} - 8\,x_{59} \\
&- 4\,x_{59}^2 + x_{60} \\
Q_{46} = &-49 - 28\,x_{60} - 4\,x_{60}^2 + x_{61} \\
Q_{47} = &-169 - 52\,x_{11} - 4\,x_{11}^2 - 8\,x_{11}\,x_{26} \\
&- 16\,x_{11}\,x_{60} - 8\,x_{11}\,x_{61} - 52\,x_{26} - 4\,x_{26}^2 \\
&- 16\,x_{26}\,x_{60} - 8\,x_{26}\,x_{61} - 104\,x_{60} - 16\,x_{60}^2 \\
&- 16\,x_{60}\,x_{61} - 52\,x_{61} - 4\,x_{61}^2 + x_{62} \\
Q_{48} = &-12 - 39\,x_{11} - 25\,x_{11}^2 - 40\,x_{11}\,x_{26} \\
&- 20\,x_{11}\,x_{62} - 32\,x_{26} - 16\,x_{26}^2 - 16\,x_{26}\,x_{62} \\
&- 16\,x_{62} - 4\,x_{62}^2 + x_{63} \\
Q_{49} = &-49 - 28\,x_{55} - 4\,x_{55}^2 + x_{64} \\
Q_{50} = &-324 - 144\,x_{55} - 16\,x_{55}^2 - 16\,x_{55}\,x_{63} \\
&- 16\,x_{55}\,x_{64} - 72\,x_{63} - 4\,x_{63}^2 - 8\,x_{63}\,x_{64} \\
&- 72\,x_{64} - 4\,x_{64}^2 + x_{65} \\
Q_{51} = &-192 - 27\,x_{12} - x_{12}^2 - 8\,x_{12}\,x_{63} \\
&- 4\,x_{12}\,x_{65} - 112\,x_{63} - 16\,x_{63}^2 - 16\,x_{63}\,x_{65} \\
&- 56\,x_{65} - 4\,x_{65}^2 + x_{66} \\
Q_{52} = &-5 - 12\,x_{10} - 4\,x_{10}^2 - 4\,x_{10}\,x_{12} \\
&- 8\,x_{10}\,x_{22} - 5\,x_{12} - x_{12}^2 - 4\,x_{12}\,x_{22} \\
&- 12\,x_{22} - 4\,x_{22}^2 + x_{67} \\
Q_{53} = &-49 - 28\,x_{10} - 4\,x_{10}^2 - 8\,x_{10}\,x_{22} \\
&- 8\,x_{10}\,x_{67} - 28\,x_{22} - 4\,x_{22}^2 - 8\,x_{22}\,x_{67} \\
&- 28\,x_{67} - 4\,x_{67}^2 + x_{68} \\
Q_{54} = &-66 - 68\,x_{10} - 16\,x_{10}^2 - 32\,x_{10}\,x_{22} \\
&- 16\,x_{10}\,x_{66} - 16\,x_{10}\,x_{68} - 68\,x_{22} - 16\,x_{22}^2 \\
&- 16\,x_{22}\,x_{66} - 16\,x_{22}\,x_{68} - 32\,x_{66} - 4\,x_{66}^2 \\
&- 8\,x_{66}\,x_{68} - 34\,x_{68} - 4\,x_{68}^2 + x_{69} \\
Q_{55} = &-1 - 4\,x_{69} - 4\,x_{69}^2 + x_{70} \\
Q_{56} = &-x_3\,x_{15} + x_{71} \\
Q_{57} = &\,x_{14} - x_{15} - 2\,x_{25}\,x_{71} - x_{72} \\
Q_{58} = &\,x_{14} - x_{15} - 4\,x_{27}\,x_{71} - 2\,x_{30}\,x_{71} \\
&- 12\,x_{71} - x_{73} \\
Q_{59} = &\,x_{14} - x_{15} - 2\,x_{32}\,x_{71} - x_{74} \\
Q_{60} = &\,x_{14} - x_{15} - 4\,x_{33}\,x_{71} - 2\,x_{35}\,x_{71} \\
&- 12\,x_{71} - x_{75}
\end{aligned}
$$

$$
\begin{aligned}
Q_{61} &= x_{14} - x_{15} - 2\,x_{37}\,x_{71} - x_{76} \\
Q_{62} &= x_{14} - x_{15} - 4\,x_{40}\,x_{71} - 2\,x_{41}\,x_{71} \\
&\quad - x_{77} \\
Q_{63} &= x_{14} - x_{15} - 2\,x_{44}\,x_{71} - x_{78} \\
Q_{64} &= x_{14} - x_{15} - 2\,x_{48}\,x_{71} - x_{79} \\
Q_{65} &= -x_{10}\,x_{11} + x_{10}\,x_{12} + x_{11}\,x_{12} - x_{12}^2 \\
&\quad + x_{80} \\
Q_{66} &= 1 + 2\,x_{13} + x_{13}^2 - x_{80}^2 \\
&\quad + x_{81} \\
Q_{67} &= -x_{81}^2 + x_{82} \\
Q_{68} &= -4\,x_{69}\,x_{71} - 2\,x_{70}\,x_{71} + x_{83} \\
Q_{69} &= -x_{14}^2 + 2\,x_{14}\,x_{15} + 2\,x_{14}\,x_{83} - x_{15}^2 \\
&\quad - 2\,x_{15}\,x_{83} - x_{83}^2 + x_{84} \\
Q_{70} &= -x_3\,x_{11} + x_{85} \\
Q_{71} &= -x_{15} - x_{15}\,x_{85} + x_{87}\,x_{88} + x_{88} \\
&\quad + x_{89} \\
Q_{72} &= -x_3\,x_{10} + x_{86} \\
Q_{73} &= -2\,x_3\,x_{10} - 2\,x_3\,x_{22} + x_{87} \\
Q_{74} &= -x_{14} - x_{14}\,x_{86} + x_{88} \\
Q_{75} &= -6\,x_3 - 6\,x_3\,x_{10} - 8\,x_3\,x_{11} - 2\,x_3\,x_{22} \\
&\quad + x_{90} \\
Q_{76} &= x_{14} + x_{14}\,x_{85} - x_{15} - x_{15}\,x_{90} \\
&\quad + x_{91} \\
Q_{77} &= -x_{14}\,x_{72} + x_{15}\,x_{72} + x_{92} \\
Q_{78} &= -x_{73}\,x_{92} + x_{93} \\
Q_{79} &= -x_{74}\,x_{93} + x_{94} \\
Q_{80} &= -x_{75}\,x_{94} + x_{95} \\
Q_{81} &= -x_{76}\,x_{95} + x_{96} \\
Q_{82} &= -x_{77}\,x_{96} + x_{97} \\
Q_{83} &= -x_{78}\,x_{97} + x_{98} \\
Q_{84} &= -x_{79}\,x_{98} + x_{99} \\
Q_{85} &= -x_{89}\,x_{99} + x_{100} \\
Q_{86} &= -x_{91}\,x_{100} + x_{101} \\
Q_{87} &= -x_{101}^2 + x_{102} \\
Q_{88} &= -x_2 - x_2\,x_{18} + x_{103} \\
Q_{89} &= -x_{16} - x_{16}\,x_{103} + x_{104} \\
Q_{90} &= -2\,x_1^2 + 2\,x_1\,x_{14} + 2\,x_1\,x_{15} + 2\,x_1\,x_{104} \\
&\quad + 2\,x_1\,x_{105} - x_{14}^2 - 2\,x_{14}\,x_{104} - x_{15}^2 \\
&\quad - 2\,x_{15}\,x_{105} + x_{20}\,x_{106} - x_{82}\,x_{102} - x_{84}\,x_{102} \\
&\quad - x_{104}^2 - x_{105}^2 \\
Q_{91} &= -x_2\,x_{17} - x_{17} - x_{17}\,x_{103} + x_{105}
\end{aligned}
$$

$$
\begin{aligned}
Q_{92} = &-31 - 9\,x_{10} - 9\,x_{11} - 4\,x_{22} \\
&- 2\,x_{25} - 4\,x_{27} - 2\,x_{30} - 2\,x_{32} \\
&- 4\,x_{33} - 2\,x_{35} - 2\,x_{37} - 4\,x_{40} \\
&- 2\,x_{41} - 2\,x_{44} - 2\,x_{48} - 4\,x_{69} \\
&- 2\,x_{70} + x_{301} - x_{304} \\
Q_{93} = &\,1 + x_{5} - x_{20}\,x_{107} + x_{111} \\
Q_{94} = &-x_{5}\,x_{110} - x_{110} - x_{110}\,x_{111} + x_{112} \\
Q_{95} = &-x_{5}\,x_{114} - x_{111}\,x_{114} - x_{114} + x_{115} \\
Q_{96} = &-x_{20}\,x_{117} + x_{119} \\
Q_{97} = &-x_{20}\,x_{127} + x_{129} \\
Q_{98} = &-x_{20}\,x_{133} + x_{135} \\
Q_{99} = &-x_{20}\,x_{109} + x_{140} \\
Q_{100} = &-x_{143}\,x_{145} + x_{146} \\
Q_{101} = &-x_{143}\,x_{146} - x_{146} + x_{147} \\
Q_{102} = &-1 + x_{144}^{2} + x_{146}^{2} - x_{147}^{2} \\
Q_{103} = &-x_{144}\,x_{149} + x_{152} \\
Q_{104} = &-1 + x_{148}^{2} + x_{149}^{2} - x_{152}^{2} \\
Q_{105} = &-4\,x_{149}\,x_{150} + x_{153} \\
Q_{106} = &\,1 - x_{144} - x_{148}\,x_{151} + x_{153} \\
Q_{107} = &-x_{144}\,x_{155} + x_{158} \\
Q_{108} = &-1 + x_{154}^{2} + x_{155}^{2} - x_{158}^{2} \\
Q_{109} = &-x_{153}\,x_{157} - x_{157} + x_{159} \\
Q_{110} = &-1 + x_{156}^{2} + x_{157}^{2} - x_{159}^{2} \\
Q_{111} = &-x_{148}\,x_{160} - x_{154} + x_{156} \\
Q_{112} = &-15 - x_{0} - 4\,x_{155}\,x_{161} + x_{157} \\
Q_{113} = &-197 - 28\,x_{0} - x_{0}^{2} - 2\,x_{0}\,x_{144} \\
&+ 2\,x_{0}\,x_{177} - 2\,x_{0}\,x_{180} - 28\,x_{144} + 2\,x_{144}\,x_{177} \\
&- 2\,x_{144}\,x_{180} - x_{162} + 28\,x_{177} - x_{177}^{2} \\
&+ 2\,x_{177}\,x_{180} - 28\,x_{180} - x_{180}^{2} \\
Q_{114} = &\,x_{0}\,x_{155} + x_{6} - x_{154} + 14\,x_{155} \\
&- x_{155}\,x_{177} + x_{155}\,x_{180} + x_{158} + x_{162}\,x_{163} \\
Q_{115} = &-x_{144}\,x_{165} + x_{168} \\
Q_{116} = &-1 + x_{164}^{2} + x_{165}^{2} - x_{168}^{2} \\
Q_{117} = &-x_{153}\,x_{167} - x_{167} + x_{169} \\
Q_{118} = &-1 + x_{166}^{2} + x_{167}^{2} - x_{169}^{2} \\
Q_{119} = &-x_{148}\,x_{170} - x_{164} + x_{166} \\
Q_{120} = &-4 - 2\,x_{1} - 2\,x_{2} - 4\,x_{165}\,x_{171} \\
&+ x_{167} - 2\,x_{190} + 2\,x_{193} \\
Q_{121} = &\,1 + x_{19}^{2} - 2\,x_{19}\,x_{144} + x_{172} \\
Q_{122} = &-x_{19}\,x_{165} + x_{20} - x_{164} + x_{168} \\
&+ x_{172}\,x_{173}
\end{aligned}
$$

$$
\begin{aligned}
Q_{123} = &-2-2\,x_{6}-x_{6}^{2}-2\,x_{6}\,x_{20}\\
&+2\,x_{6}\,x_{144}-2\,x_{20}-x_{20}^{2}+2\,x_{20}\,x_{144}\\
&+2\,x_{144}-x_{174}\\
Q_{124} = &-x_{6}\,x_{165}-x_{20}\,x_{165}+x_{107}-x_{164}\\
&-x_{165}+x_{168}+x_{174}\,x_{175}\\
Q_{125} = &-x_{144}\,x_{177}+x_{181}\\
Q_{126} = &-1+x_{176}^{2}+x_{177}^{2}-x_{181}^{2}\\
Q_{127} = &-x_{153}\,x_{179}-x_{179}+x_{182}\\
Q_{128} = &-1+x_{178}^{2}+x_{179}^{2}-x_{182}^{2}\\
Q_{129} = &-x_{148}\,x_{183}-x_{176}+x_{178}\\
Q_{130} = &-x_{177}-4\,x_{177}\,x_{184}+x_{179}+x_{180}\\
Q_{131} = &-5-4\,x_{5}-x_{5}^{2}-2\,x_{5}\,x_{111}\\
&+2\,x_{5}\,x_{144}-4\,x_{111}-x_{111}^{2}+2\,x_{111}\,x_{144}\\
&+4\,x_{144}-x_{185}\\
Q_{132} = &-x_{5}\,x_{177}+x_{108}-x_{111}\,x_{177}-x_{176}\\
&-2\,x_{177}+x_{181}+x_{185}\,x_{186}\\
Q_{133} = &-2-2\,x_{5}-x_{5}^{2}-2\,x_{5}\,x_{111}\\
&+2\,x_{5}\,x_{144}-2\,x_{111}-x_{111}^{2}+2\,x_{111}\,x_{144}\\
&+2\,x_{144}-x_{187}\\
Q_{134} = &-x_{5}\,x_{155}+x_{109}-x_{111}\,x_{155}-x_{154}\\
&-x_{155}+x_{158}+x_{187}\,x_{188}\\
Q_{135} = &-x_{144}\,x_{190}+x_{194}\\
Q_{136} = &-1+x_{189}^{2}+x_{190}^{2}-x_{194}^{2}\\
Q_{137} = &-x_{153}\,x_{192}-x_{192}+x_{195}\\
Q_{138} = &-1+x_{191}^{2}+x_{192}^{2}-x_{195}^{2}\\
Q_{139} = &-x_{148}\,x_{196}-x_{189}+x_{191}\\
Q_{140} = &-x_{190}-4\,x_{190}\,x_{197}+x_{192}+x_{193}\\
Q_{141} = &-x_{5}\,x_{190}-x_{111}\,x_{190}+x_{113}+x_{185}\,x_{198}\\
&-x_{189}-2\,x_{190}+x_{194}\\
Q_{142} = &-x_{144}\,x_{200}+x_{203}\\
Q_{143} = &-1+x_{199}^{2}+x_{200}^{2}-x_{203}^{2}\\
Q_{144} = &-x_{153}\,x_{202}-x_{202}+x_{204}\\
Q_{145} = &-1+x_{201}^{2}+x_{202}^{2}-x_{204}^{2}\\
Q_{146} = &-x_{148}\,x_{205}-x_{199}+x_{201}\\
Q_{147} = &-x_{10}-4\,x_{200}\,x_{206}+x_{202}\\
Q_{148} = &-x_{5}\,x_{200}-x_{111}\,x_{200}+x_{116}+x_{185}\,x_{207}\\
&-x_{199}-2\,x_{200}+x_{203}\\
Q_{149} = &\,1-x_{144}^{2}+x_{208}\\
Q_{150} = &-x_{5}\,x_{154}-x_{5}\,x_{158}-x_{111}\,x_{154}-x_{111}\,x_{158}\\
&+x_{117}+x_{144}\,x_{158}-x_{154}-x_{155}\,x_{208}\\
&-x_{158}+x_{187}\,x_{209}\\
Q_{151} = &-x_{144}\,x_{211}+x_{214}\\
Q_{152} = &-1+x_{210}^{2}+x_{211}^{2}-x_{214}^{2}
\end{aligned}
$$

$$
\begin{aligned}
Q_{153} &= -x_{153}\,x_{213} - x_{213} + x_{215}\\
Q_{154} &= -1 + x_{212}^2 + x_{213}^2 - x_{215}^2\\
Q_{155} &= -x_{148}\,x_{216} - x_{210} + x_{212}\\
Q_{156} &= -x_{11} - 4\,x_{211}\,x_{217} + x_{213}\\
Q_{157} &= -x_{5}\,x_{211} - x_{111}\,x_{211} + x_{120} + x_{185}\,x_{218}\\
&\quad - x_{210} - 2\,x_{211} + x_{214}\\
Q_{158} &= -x_{144}\,x_{220} + x_{223}\\
Q_{159} &= -1 + x_{219}^2 + x_{220}^2 - x_{223}^2\\
Q_{160} &= -x_{153}\,x_{222} - x_{222} + x_{224}\\
Q_{161} &= -1 + x_{221}^2 + x_{222}^2 - x_{224}^2\\
Q_{162} &= -x_{148}\,x_{225} - x_{219} + x_{221}\\
Q_{163} &= -x_{12} - 4\,x_{220}\,x_{226} + x_{222}\\
Q_{164} &= -x_{5}\,x_{220} - x_{111}\,x_{220} + x_{122} + x_{185}\,x_{227}\\
&\quad - x_{219} - 2\,x_{220} + x_{223}\\
Q_{165} &= -x_{144}\,x_{229} + x_{232}\\
Q_{166} &= -1 + x_{228}^2 + x_{229}^2 - x_{232}^2\\
Q_{167} &= -x_{153}\,x_{231} - x_{231} + x_{233}\\
Q_{168} &= -1 + x_{230}^2 + x_{231}^2 - x_{233}^2\\
Q_{169} &= -x_{148}\,x_{234} - x_{228} + x_{230}\\
Q_{170} &= -x_{13} - 4\,x_{229}\,x_{235} + x_{231}\\
Q_{171} &= -x_{5}\,x_{229} - x_{111}\,x_{229} + x_{124} + x_{185}\,x_{236}\\
&\quad - x_{228} - 2\,x_{229} + x_{232}\\
Q_{172} &= -x_{144}\,x_{238} + x_{241}\\
Q_{173} &= -1 + x_{237}^2 + x_{238}^2 - x_{241}^2\\
Q_{174} &= -x_{153}\,x_{240} - x_{240} + x_{242}\\
Q_{175} &= -1 + x_{239}^2 + x_{240}^2 - x_{242}^2\\
Q_{176} &= -x_{148}\,x_{243} - x_{237} + x_{239}\\
Q_{177} &= -x_{14} - 4\,x_{238}\,x_{244} + x_{240}\\
Q_{178} &= -x_{5}\,x_{238} - x_{111}\,x_{238} + x_{126} + x_{185}\,x_{245}\\
&\quad - x_{237} - 2\,x_{238} + x_{241}\\
Q_{179} &= -x_{144}\,x_{247} + x_{250}\\
Q_{180} &= -1 + x_{246}^2 + x_{247}^2 - x_{250}^2\\
Q_{181} &= -x_{153}\,x_{249} - x_{249} + x_{251}\\
Q_{182} &= -1 + x_{248}^2 + x_{249}^2 - x_{251}^2\\
Q_{183} &= -x_{148}\,x_{252} - x_{246} + x_{248}\\
Q_{184} &= -1 - x_{2} - 4\,x_{247}\,x_{253} + x_{249}\\
Q_{185} &= -x_{5}\,x_{247} - x_{111}\,x_{247} + x_{127} + x_{187}\,x_{254}\\
&\quad - x_{246} - x_{247} + x_{250}\\
Q_{186} &= -x_{144}\,x_{256} + x_{259}\\
Q_{187} &= -1 + x_{255}^2 + x_{256}^2 - x_{259}^2\\
Q_{188} &= -x_{153}\,x_{258} - x_{258} + x_{260}\\
Q_{189} &= -1 + x_{257}^2 + x_{258}^2 - x_{260}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{190} &= -x_{148}\,x_{261} - x_{255} + x_{257}\\
Q_{191} &= -x_{15} - 4\,x_{256}\,x_{262} + x_{258}\\
Q_{192} &= -x_{5}\,x_{256} - x_{111}\,x_{256} + x_{130} + x_{185}\,x_{263}\\
&\quad - x_{255} - 2\,x_{256} + x_{259}\\
Q_{193} &= -x_{144}\,x_{265} + x_{268}\\
Q_{194} &= -1 + x_{264}^{2} + x_{265}^{2} - x_{268}^{2}\\
Q_{195} &= -x_{153}\,x_{267} - x_{267} + x_{269}\\
Q_{196} &= -1 + x_{266}^{2} + x_{267}^{2} - x_{269}^{2}\\
Q_{197} &= -x_{148}\,x_{270} - x_{264} + x_{266}\\
Q_{198} &= -x_{16} - 4\,x_{265}\,x_{271} + x_{267}\\
Q_{199} &= -x_{5}\,x_{265} - x_{111}\,x_{265} + x_{132} + x_{185}\,x_{272}\\
&\quad - x_{264} - 2\,x_{265} + x_{268}\\
Q_{200} &= -x_{144}\,x_{274} + x_{277}\\
Q_{201} &= -1 + x_{273}^{2} + x_{274}^{2} - x_{277}^{2}\\
Q_{202} &= -x_{153}\,x_{276} - x_{276} + x_{278}\\
Q_{203} &= -1 + x_{275}^{2} + x_{276}^{2} - x_{278}^{2}\\
Q_{204} &= -x_{148}\,x_{279} - x_{273} + x_{275}\\
Q_{205} &= -1 - x_{1} - 4\,x_{274}\,x_{280} + x_{276}\\
Q_{206} &= -x_{5}\,x_{274} - x_{111}\,x_{274} + x_{133} + x_{187}\,x_{281}\\
&\quad - x_{273} - x_{274} + x_{277}\\
Q_{207} &= -x_{144}\,x_{283} + x_{286}\\
Q_{208} &= -1 + x_{282}^{2} + x_{283}^{2} - x_{286}^{2}\\
Q_{209} &= -x_{153}\,x_{285} - x_{285} + x_{287}\\
Q_{210} &= -1 + x_{284}^{2} + x_{285}^{2} - x_{287}^{2}\\
Q_{211} &= -x_{148}\,x_{288} - x_{282} + x_{284}\\
Q_{212} &= -x_{17} - 4\,x_{283}\,x_{289} + x_{285}\\
Q_{213} &= -x_{5}\,x_{283} - x_{111}\,x_{283} + x_{136} + x_{185}\,x_{290}\\
&\quad - x_{282} - 2\,x_{283} + x_{286}\\
Q_{214} &= -x_{144}\,x_{292} + x_{295}\\
Q_{215} &= -1 + x_{291}^{2} + x_{292}^{2} - x_{295}^{2}\\
Q_{216} &= -x_{153}\,x_{294} - x_{294} + x_{296}\\
Q_{217} &= -1 + x_{293}^{2} + x_{294}^{2} - x_{296}^{2}\\
Q_{218} &= -x_{148}\,x_{297} - x_{291} + x_{293}\\
Q_{219} &= -x_{18} - 4\,x_{292}\,x_{298} + x_{294}\\
Q_{220} &= -x_{5}\,x_{292} - x_{111}\,x_{292} + x_{138} + x_{185}\,x_{299}\\
&\quad - x_{291} - 2\,x_{292} + x_{295}\\
Q_{221} &= -x_{144}\,x_{301} + x_{305}\\
Q_{222} &= -1 + x_{300}^{2} + x_{301}^{2} - x_{305}^{2}\\
Q_{223} &= -x_{153}\,x_{303} - x_{303} + x_{306}\\
Q_{224} &= -1 + x_{302}^{2} + x_{303}^{2} - x_{306}^{2}\\
Q_{225} &= -x_{148}\,x_{307} - x_{300} + x_{302}\\
Q_{226} &= -x_{301} - 4\,x_{301}\,x_{308} + x_{303} + x_{304}
\end{aligned}
$$

$$
\begin{aligned}
Q_{227} = {} & - x_{5}\,x_{301} - x_{111}\,x_{301} + x_{141} + x_{185}\,x_{309} \\
& - x_{300} - 2\,x_{301} + x_{305} \\
Q_{228} = {} & - x_{155}\,x_{165} + x_{310} \\
Q_{229} = {} & - x_{177}\,x_{310} + x_{311} \\
Q_{230} = {} & - x_{190}\,x_{311} + x_{312} \\
Q_{231} = {} & - x_{200}\,x_{312} + x_{313} \\
Q_{232} = {} & - x_{211}\,x_{313} + x_{314} \\
Q_{233} = {} & - x_{220}\,x_{314} + x_{315} \\
Q_{234} = {} & - x_{229}\,x_{315} + x_{316} \\
Q_{235} = {} & - x_{238}\,x_{316} + x_{317} \\
Q_{236} = {} & - x_{247}\,x_{317} + x_{318} \\
Q_{237} = {} & - x_{256}\,x_{318} + x_{319} \\
Q_{238} = {} & - x_{265}\,x_{319} + x_{320} \\
Q_{239} = {} & - x_{274}\,x_{320} + x_{321} \\
Q_{240} = {} & - x_{283}\,x_{321} + x_{322} \\
Q_{241} = {} & - x_{292}\,x_{322} + x_{323} \\
Q_{242} = {} & - x_{301}\,x_{323} + x_{324} \\
Q_{243} = {} & - x_{324}^{2} + x_{325} \\
Q_{244} = {} & x_{149} - x_{325}\,x_{326} \\
Q_{245} = {} & - 703 - 90\,x_{0} - 3\,x_{0}^{2} + 2\,x_{0}\,x_{177} \\
& - 2\,x_{0}\,x_{180} - 18\,x_{1} - 5\,x_{1}^{2} - 8\,x_{1}\,x_{2} \\
& - 8\,x_{1}\,x_{190} + 8\,x_{1}\,x_{193} - 18\,x_{2} - 5\,x_{2}^{2} \\
& - 8\,x_{2}\,x_{190} + 8\,x_{2}\,x_{193} - 6\,x_{5} - 2\,x_{5}^{2} \\
& - 4\,x_{5}\,x_{111} - 2\,x_{6} - 2\,x_{6}^{2} - 2\,x_{6}\,x_{20} \\
& - x_{10}^{2} - x_{11}^{2} - x_{12}^{2} - x_{13}^{2} \\
& - x_{14}^{2} - x_{15}^{2} - x_{16}^{2} - x_{17}^{2} \\
& - x_{18}^{2} - x_{19}^{2} - 2\,x_{20} - 2\,x_{20}^{2} \\
& - x_{107}^{2} - x_{108}^{2} - x_{109}^{2} - 6\,x_{111} \\
& - 2\,x_{111}^{2} - x_{113}^{2} - x_{116}^{2} - x_{117}^{2} \\
& - x_{120}^{2} - x_{122}^{2} - x_{124}^{2} - x_{126}^{2} \\
& - x_{127}^{2} - x_{130}^{2} - x_{132}^{2} - x_{133}^{2} \\
& - x_{136}^{2} - x_{138}^{2} - x_{141}^{2} + x_{143} \\
& + 28\,x_{177} - 2\,x_{177}^{2} + 4\,x_{177}\,x_{180} - 28\,x_{180} \\
& - 2\,x_{180}^{2} - 16\,x_{190} - 5\,x_{190}^{2} + 10\,x_{190}\,x_{193} \\
& + 16\,x_{193} - 5\,x_{193}^{2} - x_{301}^{2} + 2\,x_{301}\,x_{304} \\
& - x_{304}^{2} \\
Q_{246} = {} & - 1 + x_{0} - x_{507}^{2} \\
Q_{247} = {} & x_{1} - x_{327}^{2} - x_{328}^{2} - x_{329} \\
& - x_{329}^{2} \\
Q_{248} = {} & - 1 - x_{3}\,x_{5} + x_{5} + x_{6} \\
& - x_{330}^{2} - x_{331}^{2} - x_{332} - x_{332}^{2} \\
Q_{249} = {} & x_{3}\,x_{5} - x_{6} - x_{333}^{2} - x_{334}^{2} \\
& - x_{335} - x_{335}^{2} \\
Q_{250} = {} & x_{2} + x_{6} - x_{6}^{2}
\end{aligned}
$$

$$
\begin{aligned}
Q_{251} = {} & -1 + x_{10} - x_{336}^2 - x_{337}^2 \\
& - x_{338} - x_{338}^2 \\
Q_{252} = {} & -1 + x_{11} - x_{339}^2 - x_{340}^2 \\
& - x_{341} - x_{341}^2 \\
Q_{253} = {} & -1 + x_{12} - x_{342}^2 - x_{343}^2 \\
& - x_{344} - x_{344}^2 \\
Q_{254} = {} & -1 + x_{13} - x_{345}^2 - x_{346}^2 \\
& - x_{347} - x_{347}^2 \\
Q_{255} = {} & -1 + x_{14} - x_{348}^2 - x_{349}^2 \\
& - x_{350} - x_{350}^2 \\
Q_{256} = {} & -1 + x_{15} - x_{351}^2 - x_{352}^2 \\
& - x_{353} - x_{353}^2 \\
Q_{257} = {} & -1 + x_{16} - x_{354}^2 - x_{355}^2 \\
& - x_{356} - x_{356}^2 \\
Q_{258} = {} & -1 + x_{17} - x_{357}^2 - x_{358}^2 \\
& - x_{359} - x_{359}^2 \\
Q_{259} = {} & -1 + x_{18} - x_{360}^2 - x_{361}^2 \\
& - x_{362} - x_{362}^2 \\
Q_{260} = {} & x_{19} - x_{363}^2 - x_{364}^2 - x_{365} \\
& - x_{365}^2 \\
Q_{261} = {} & - x_{10} - x_{11} - x_{12} - x_{13} \\
& - x_{14} - x_{15} - x_{16} - x_{17} \\
& - x_{18} + 9\,x_{20}^2 - x_{366}^2 - x_{367}^2 \\
& - x_{368} - x_{368}^2 \\
Q_{262} = {} & - x_5\,x_{109} + x_{108} - x_{109}\,x_{112} - x_{369}^2 \\
& - x_{370}^2 - x_{371} - x_{371}^2 \\
Q_{263} = {} & -1 + x_5\,x_{109} - x_{108} + x_{109} \\
& + x_{109}\,x_{112} - x_{372}^2 - x_{373}^2 - x_{374} \\
& - x_{374}^2 \\
Q_{264} = {} & x_{111} - x_{375}^2 - x_{376}^2 - x_{377} \\
& - x_{377}^2 \\
Q_{265} = {} & - x_{19}\,x_{109} - x_{109}\,x_{115} + x_{113} - x_{378}^2 \\
& - x_{379}^2 - x_{380} - x_{380}^2 \\
Q_{266} = {} & -1 + x_{19}\,x_{109} + x_{109} + x_{109}\,x_{115} \\
& - x_{113} - x_{381}^2 - x_{382}^2 - x_{383} \\
& - x_{383}^2 \\
Q_{267} = {} & x_5 - x_{19} + x_{111} - x_{384}^2 \\
& - x_{385}^2 - x_{386} - x_{386}^2 \\
Q_{268} = {} & x_{116} - x_{118}\,x_{119} - x_{387}^2 - x_{388}^2 \\
& - x_{389} - x_{389}^2 \\
Q_{269} = {} & -1 - x_{116} + x_{117} + x_{118}\,x_{119} \\
& - x_{390}^2 - x_{391}^2 - x_{392} - x_{392}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{270} = &- x_{119}\,x_{121} + x_{120} - x_{393}^2 - x_{394}^2 \\
&- x_{395} - x_{395}^2 \\
Q_{271} = &- 1 + x_{117} + x_{119}\,x_{121} - x_{120} \\
&- x_{396}^2 - x_{397}^2 - x_{398} - x_{398}^2 \\
Q_{272} = &- x_{119}\,x_{123} + x_{122} - x_{399}^2 - x_{400}^2 \\
&- x_{401} - x_{401}^2 \\
Q_{273} = &- 1 + x_{117} + x_{119}\,x_{123} - x_{122} \\
&- x_{402}^2 - x_{403}^2 - x_{404} - x_{404}^2 \\
Q_{274} = &- x_{119}\,x_{125} + x_{124} - x_{405}^2 - x_{406}^2 \\
&- x_{407} - x_{407}^2 \\
Q_{275} = &- 1 + x_{117} + x_{119}\,x_{125} - x_{124} \\
&- x_{408}^2 - x_{409}^2 - x_{410} - x_{410}^2 \\
Q_{276} = &\, x_{126} - x_{128}\,x_{129} - x_{411}^2 - x_{412}^2 \\
&- x_{413} - x_{413}^2 \\
Q_{277} = &- 1 - x_{126} + x_{127} + x_{128}\,x_{129} \\
&- x_{414}^2 - x_{415}^2 - x_{416} - x_{416}^2 \\
Q_{278} = &- x_{129}\,x_{131} + x_{130} - x_{417}^2 - x_{418}^2 \\
&- x_{419} - x_{419}^2 \\
Q_{279} = &- 1 + x_{127} + x_{129}\,x_{131} - x_{130} \\
&- x_{420}^2 - x_{421}^2 - x_{422} - x_{422}^2 \\
Q_{280} = &\, x_{132} - x_{134}\,x_{135} - x_{423}^2 - x_{424}^2 \\
&- x_{425} - x_{425}^2 \\
Q_{281} = &- 1 - x_{132} + x_{133} + x_{134}\,x_{135} \\
&- x_{426}^2 - x_{427}^2 - x_{428} - x_{428}^2 \\
Q_{282} = &- x_{135}\,x_{137} + x_{136} - x_{429}^2 - x_{430}^2 \\
&- x_{431} - x_{431}^2 \\
Q_{283} = &- 1 + x_{133} + x_{135}\,x_{137} - x_{136} \\
&- x_{432}^2 - x_{433}^2 - x_{434} - x_{434}^2 \\
Q_{284} = &\, x_{138} - x_{139}\,x_{140} - x_{435}^2 - x_{436}^2 \\
&- x_{437} - x_{437}^2 \\
Q_{285} = &- 1 + x_{109} - x_{138} + x_{139}\,x_{140} \\
&- x_{438}^2 - x_{439}^2 - x_{440} - x_{440}^2 \\
Q_{286} = &- x_{119}\,x_{142} + x_{141} - x_{441}^2 - x_{442}^2 \\
&- x_{443} - x_{443}^2 \\
Q_{287} = &- 1 + x_{117} + x_{119}\,x_{142} - x_{141} \\
&- x_{444}^2 - x_{445}^2 - x_{446} - x_{446}^2 \\
Q_{288} = &\, x_{144} - x_{447}^2 - x_{448}^2 - x_{449} \\
&- x_{449}^2 \\
Q_{289} = &- 1 + x_{145} - x_{450}^2 - x_{451}^2 \\
&- x_{452} - x_{452}^2 \\
Q_{290} = &- 1 + x_{150} - x_{453}^2 - x_{454}^2 \\
&- x_{455} - x_{455}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{291} = {} & -15 - x_{0} + x_{155} - x_{456}^{2} \\
& - x_{457}^{2} - x_{458} - x_{458}^{2} \\
Q_{292} = {} & -4 - 2\,x_{1} - 2\,x_{2} + x_{165} \\
& - 2\,x_{190} + 2\,x_{193} - x_{459}^{2} - x_{460}^{2} \\
& - x_{461} - x_{461}^{2} \\
Q_{293} = {} & x_{180} - x_{462}^{2} - x_{463}^{2} - x_{464} \\
& - x_{464}^{2} \\
Q_{294} = {} & x_{193} - x_{465}^{2} - x_{466}^{2} - x_{467} \\
& - x_{467}^{2} \\
Q_{295} = {} & -x_{10} + x_{200} - x_{468}^{2} - x_{469}^{2} \\
& - x_{470} - x_{470}^{2} \\
Q_{296} = {} & -x_{11} + x_{211} - x_{471}^{2} - x_{472}^{2} \\
& - x_{473} - x_{473}^{2} \\
Q_{297} = {} & -x_{12} + x_{220} - x_{474}^{2} - x_{475}^{2} \\
& - x_{476} - x_{476}^{2} \\
Q_{298} = {} & -x_{13} + x_{229} - x_{477}^{2} - x_{478}^{2} \\
& - x_{479} - x_{479}^{2} \\
Q_{299} = {} & -x_{14} + x_{238} - x_{480}^{2} - x_{481}^{2} \\
& - x_{482} - x_{482}^{2} \\
Q_{300} = {} & -1 - x_{2} + x_{247} - x_{483}^{2} \\
& - x_{484}^{2} - x_{485} - x_{485}^{2} \\
Q_{301} = {} & -x_{15} + x_{256} - x_{486}^{2} - x_{487}^{2} \\
& - x_{488} - x_{488}^{2} \\
Q_{302} = {} & -x_{16} + x_{265} - x_{489}^{2} - x_{490}^{2} \\
& - x_{491} - x_{491}^{2} \\
Q_{303} = {} & -1 - x_{1} + x_{274} - x_{492}^{2} \\
& - x_{493}^{2} - x_{494} - x_{494}^{2} \\
Q_{304} = {} & -x_{17} + x_{283} - x_{495}^{2} - x_{496}^{2} \\
& - x_{497} - x_{497}^{2} \\
Q_{305} = {} & -x_{18} + x_{292} - x_{498}^{2} - x_{499}^{2} \\
& - x_{500} - x_{500}^{2} \\
Q_{306} = {} & x_{304} - x_{501}^{2} - x_{502}^{2} - x_{503} \\
& - x_{503}^{2} \\
Q_{307} = {} & -1 + x_{326} - x_{504}^{2} - x_{505}^{2} \\
& - x_{506} - x_{506}^{2}
\end{aligned}
$$